

John Huynh

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EDUCATION

University of California, San Diego

San Diego, CA

Bachelor of Science in Mathematics and Computer Science

Sept. 2022 – Mar. 2026

- **Relevant Coursework:** Software Engineering, Data Structures, Algorithms, Machine Learning, Recommender Systems, Systems Programming, Theory of Computation

EXPERIENCE

Software Engineering Intern

San Diego, CA

Resmed

Jun. 2025 – Present

- Developed a data-driven engineering metrics platform using React, TypeScript, Spring Boot, and MongoDB to visualize review quality, response times, and approval patterns, enabling leadership to identify bottlenecks and improve code review practices with AI-powered pull request summarization.
- Automated release documentation for IoT microservices by building TypeScript tooling with MCP-based workflows and internal APIs, eliminating manual effort per release.
- Improved observability across 20+ IoT services by consolidating Datadog monitors into reusable Terraform templates and adding 12 new monitors for data ingestion endpoints.
- Refactored IoT data ingestion deployments from per-endpoint to grouped deployments, reducing deployment complexity by restructuring Kubernetes ConfigMaps, YAML deployment manifests, GitHub Actions build matrices, and gateway routing.

PROJECTS

ReleaseOps Backend | *Java, Spring Boot, PostgreSQL, Docker*

- Building a backend release management system for tracking services, releases, readiness checks, deployment status, and documentation links through RESTful APIs.
- Designing a layered Spring Boot architecture with controllers, services, repositories, DTOs, validation, and centralized exception handling.
- Planning to integrate PostgreSQL persistence, Flyway database migrations, Dockerized local development, and automated testing with JUnit and Mockito.

Habitizer | *Android Studio, Java*

- Developed a mobile routine management app with dynamic task lists, integrated timers, and customizable routine tracking workflows.
- Applied the MVP design pattern to create a modular architecture with clearer separation between UI logic, business logic, and application state.
- Implemented behavior-driven scenarios, unit tests, and story-based acceptance tests to validate timer behavior and end-to-end routine workflows.
- Integrated GitHub Actions for continuous integration and automated test execution, supporting a team workflow with protected branches and code review.

Anime Data Analysis and Rating Prediction System | *Python, NumPy, Pandas, Matplotlib, Scikit-learn*

- Analyzed anime metadata and rating patterns using Python-based data pipelines and visualizations to identify trends in a large dataset.
- Built a rating prediction model with RandomForestRegressor, achieving an R^2 score of 0.74.
- Improved model performance through hyperparameter tuning with RandomizedSearchCV and evaluated feature importance to identify the strongest predictors.

Data Compressor | *C++*

- Implemented a file compression system using Huffman coding to reduce file sizes by up to 30%.
- Built supporting data structures and optimized compression logic for improved performance on larger inputs.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, C++, SQL, HTML/CSS

Frameworks/Technologies: Spring Boot, React, Node.js, Express, JUnit, REST APIs

Cloud/Tools: AWS, Docker, Kubernetes, Terraform, Datadog, Git, GitHub Actions, CI/CD

Databases: PostgreSQL, MongoDB, SQLite